The POISSON PROCESS

Dean Cochran

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1 Introduction

In real life, often we are interested in multiple observations over a period of time. For these cases we are interested in random processes. A random process is a collection of random variables usually indexed by time (t). This is why random processes are random functions of time. In general, when we have a random process N(t), where t can take on any real values in an interval on the real line, then N(t) is a continuous-time random process. A discrete-time random process is just a sequence of random variables. For this reason, we can refer to this process as random sequences.

Definition 1.1. Continuous and Discrete Random Processes
A continuous-time random process is a random process X(t),t$\in J$, where J is an interval on the real line such as [-1,1], [0,∞), (-∞,∞), etc.

A discrete-time random process (or a random sequence) is a random process X(n)=In,n$\in J$, where J is a countable set such as N(Natural numbers) or Z (set of all integers).

In this chapter, we will focus on some specific random processes that are used frequently in applications. More specifically, we will discuss the Poisson process

2 Counting Processes

For certain scenarios we count the number of occurrences of some type of event. In these scenarios we are dealing with counting processes.

Definition 2.1. Counting Process
A random process N(t),t$\in [0, \infty]$ is said to be a counting process if N(t) is the number of events occurred from time 0 up to and including time t. For a counting process, we assume

1. N(0)=0
2. N(t)$\in [0, 1, 2, \cdots]$, for all t$\in [0, \infty]$;
3. for 0$\leq$s<t, N(t)-N(s) shows the number of events that occur in the interval (s,t].

Counting processes model arrivals refer to the occurrence of each event as an "arrival". For example, if N(t) is the number of cars in a city that pass a certain intersection up to time t, we still refer to each passing car as an arrival.

Notice that the only randomness in this process consists of the variable arrival times.

Definition 2.2. Independent Increments
Let N(t),t$\in [0, \infty]$ be a continuous-time random process. We say that a counting process has independent increments if and only if,

for all

$$0 \leq t < t_2 < t_3 \cdots < t_n \tag{1}$$

the random variables

$$N(t_2) - N(t_1), N(t_3) - N(t2), \cdots, N(t_n) - N(t_{n-1)} \tag{2}$$

are independent.

Any counting process, N$(t_i) - N(t_{i-1})$ is the amount of arrivals in the interval from $(t_{i-1}, t_i]$. This means that counting processes have independent increments when the arrivals in non-overlapping intervals $(t_{i-1}, t_i]$ are independent.

When dealing with independent increments and intervals the analysis for counting processes are simplified and made more efficient.

3 The Basic Concepts of the Poisson Process

This is commonly applied counting processes out there. It can be observed that is commonly used when counting the arrivals of certain events that happen at a certain rate, but are still completely random.

Here we briefly review the properties of a Poisson Random Variable.

Definition 3.1. Poisson Random Variable
X is said to be a Poisson Random variable with parameter μ, if its range is $R_X = 0, 1, 2, 3, \cdots$, and its PMF is given by

$$P_X(k) = \begin{cases} \frac{e^{-\mu}\mu^k}{k!} & \text{for } k \in R_X \\ 0 & \text{otherwise} \end{cases}$$

1. E(X)=μ, and Var(X)=μ
2. If X_i has a Poisson distribution with parameter(μ_i), for i=1,2,$\cdots$,n, and X_i is independent, then

$$X_1 + X_2 + \cdots + X_n \text{ has a Poisson distribution with parameter} \quad (3)$$

$$(\mu_1 + \mu_2 + \cdots + \mu_n). \quad (4)$$

3. The Poisson distribution can be viewed as the limit of binomial distribution.

Theorem 3.1. *Let Y_n have a Binomial distribution with parameter (n,p(n)). Let $\mu > 0$ be a fixed real number, and $\lim_{n\to\infty} np = \mu$. Then, the PMF of Y_n converges to a Poisson distribution with parameter (μ), as $n \to \infty$. That is, for any $k \in 0, 1, 2, \cdots$, we have*

$$\lim_{n\to\infty} P_{Y_n}(k) = \frac{e^{-\mu}\mu^k}{k!}.$$

3.1 Poisson Process as the Limit of a Bernoulli Process

After understanding the basis of the Poisson Random Variable, assume events occur completely at random at rate of λ. Assuming that at the start of the observation (t=0), no arrivals have occurred. Therefore N(t), when t=0, is equal to 0. Knowing that in between each arrival there is a variable length δ, we can divide all intervals to create indefinitely small sub-intervals by which we can count. Thus, the intervals are $(0, \delta], (\delta, 2\delta], (2\delta, 3\delta], \cdots, ((k-1)\delta, k\delta]$. More generally, the kth interval is $((k-1)\delta, k\delta]$.

For example, assume we flip a coin for every interval. With this we can say that the probability of observing heads on any given interval is equivalent to $\lambda\delta$. Here, we have an arrival at time $t = k\delta$, if the kth coin flip results in a heads.

With this being said we can consider N(t) to be defined as the number of heads from 0 to t. Since there are n time slots in the interval from (0,t] since $n = 1/\delta$. We can now reiterate and clarify that N(t) is the number of heads in n coin flips. Therefore N(t) has a Binomial distribution with parameter (n,p).

Notice that $p = \lambda\delta$.

$$\begin{aligned} np &= n\lambda\delta \\ &= \frac{t}{\delta} \cdot \lambda\delta \\ &= \lambda t. \end{aligned}$$

As $\delta \to 0$, the PMF of N(t) converges to a Poisson distribution with rate λt.

The development of this conclusion can be very mathematically complex, but the resulting random process is what we can call a Poisson Process with a rate λ.

3.2 The First Definition of the Poisson Process

Definition 3.2. 1st Definition of the Poisson Process
Let $\lambda > 0$ be fixed. The counting process $N(t), t \in [0, \infty)$ is called a Poisson process with rate λ if all the following conditions hold:

1. N(0)=0
2. N(t) has independent increments
3. the number of arrivals in any interval of length $t > 0$ has Poisson(λt) distribution.

Example 3.1. The number of cars that pass an intersection can be modeled by a Poisson process with intensity λ=10 cars per hour.

1. Find the probability that there are 2 car arrivals between 10:00am and 10:20am.
2. Find the probability that there are 3 car arrivals between 10:00am and 10:20am and 7 customers between 10:20am and 11:00am. ‘

Solutions$\cdots$

1. Here, λ=10 and the interval between 10:00am and 10:20am has length t=1/3 hours. Thus, if X is the number of arrivals in that interval, we can write X has a Poisson distribution with parameter(10/3). Therefore,

$$\begin{aligned} P(X = 2) &= \frac{e^{-\frac{10}{3}} \left(\frac{10}{3}\right)^2}{2!} \\ &\approx 0.2 \end{aligned}$$

2. Here, we have two non-overlapping intervals I_1=(10:00 a.m., 10:20 a.m.] and I_2= (10:20 a.m., 11 a.m.]. Thus, we can say that the probability of observing

3 arrivals in I_1 and observing 7 arrivals in I_2 is equivalent to the probability of observing 3 arrivals in I_1 multiplied by probability of observing 7 arrivals in I_2. Since the lengths of the intervals are t_1=1/3 and t_2=2/3 respectively, we obtain λ_{t_1}=10/3 and λ_{t_2}=20/3. Thus, we can say that the probability of observing 3 arrivals in I_1 and observing 7 arrivals in I_2 is equivalent to

$$= \frac{e^{-\frac{10}{3}}\left(\frac{10}{3}\right)^3}{3!} \cdot \frac{e^{-\frac{20}{3}}\left(\frac{20}{3}\right)^7}{7!}$$
$$\approx 0.0325$$

3.3 The Second Definition of the Poisson Process

Let N(δ) be a Poisson Process with a rate λ. When considering the arrivals in a given length of time, consider an extremely short interval of length δ. Then the total amount of arrivals in this particular interval has the same distribution as N(δ). In particular, we can write,

$$\begin{aligned} P(N(\delta) = 0) &= e^{-\lambda\delta} \\ &= 1 - \lambda\delta + \frac{\lambda^2}{2}\delta^2 - \cdots \text{ (Taylor Series).} \end{aligned}$$

Note: If t is small, the terms that include second or higher powers of δ are negligible so we can write: $P(N(t) = 0) = 1 - \lambda\delta + O(\delta)$

More specifically O(δ) means that,

$$\lim_{\delta \to 0} \frac{g(\delta)}{\delta} = 0.$$

Therefore,

$$\begin{aligned} P(N(\delta) = 1) &= e^{-\lambda\delta}\lambda\delta \\ &= \lambda\delta\left(1 - \lambda\delta + \frac{\lambda^2}{2}\delta^2 - \cdots\right) \text{ (Taylor Series)} \\ &= \lambda\delta + \left(-\lambda^2\delta^2 + \frac{\lambda^3}{2}\delta^3 \cdots\right) \\ &= \lambda\delta + O(\delta). \end{aligned}$$

The equations above lead us to the Second Definition of the Poisson Process that is discussed in this chapter.

Definition 3.3. 2nd Definition of the Poisson Process
Let $\lambda > 0$ be fixed. The counting process $N(t), t \in [0, \infty)$ is called a Poisson Process with rate λ if all the following conditions hold:

1. N(0)=0
2. N(δ) has independent increments
3. P(N(δ)=0)=1-$\lambda\delta$+)O(δ)
   P(N(δ)=1)=$\lambda\delta$+O(δ)
   P(N(δ) $\geq$2)=O(δ)

4 Arrival and Inter-arrival Times

From what we have established, if N(t) is a Poisson Process with rate λ. Let X_1 be the time of the first arrival. Therefore,

$$\begin{aligned} P(X_1 > t) &= P\big(\text{no arrival in } (0, t]\big) \\ &= e^{-\lambda t}. \end{aligned}$$

From this we can conclude the cumulative distribution function to be F_X(X)=1 - $e^{-\lambda t}$

$$F_{X_1}(t) = \begin{cases} 1 - e^{-\lambda t} & t > 0 \\ 0 & \text{otherwise} \end{cases}$$

Therefore, X_1 has an Exponential distribution with parameter λ. Note that the two intervals (0, s] and [s, s+t] are independent when $s > 0$ and $t > 0$. We can write

$$\begin{aligned} P(X_2 > t | X_1 = s) &= P\big(\text{no arrival in } (s, s+t] | X_1 = s\big) \\ &= e^{-\lambda t}. \end{aligned}$$

This supports that X_2 has an Exponential distribution with parameter λ, and that X_1 and X_2 are both inherently independent. Similarly, all X_i's are independent and X_i has an Exponential distribution with parameter λ for i=1,2,3,$\cdots$.

Definition 4.1. Inter-Arrival times for Poisson Process
If N(t) is a Poisson process with rate λ, then the inter-arrival times X_1, X_2, $\cdots$ are independent and X_i has an Exponential distribution with parameter λ, for i=1,2,3,$\cdots$.

Remember that if X is Exponential with parameter λ> 0, then X is a memory less random variable, that is,

$$P(X > x + a|X > a) = P(X > x), \text{for a} x \geq 0. \tag{5}$$

Example 4.1.

Now that the distribution of inter-arrival times is defined for all Xi's for n = 1,2,3,... The distribution of arrival times can now be defined,

$$T_1 = X_1 \tag{6}$$
$$T_2 = X_1 + X_2 \tag{7}$$
$$T_3 = X_1 + X_2 + X_3 \tag{8}$$

We can define T_n as the sum of n independent Exponential random variables with parameter λ.

Theorem 4.1. *Suppose Y has a Gamma(a,b) distribution and X has an Exponential(b) distribution are independent, so that*

$$f_Y(y) = \frac{b^a y^{a-1} e^{-by}}{\Gamma(a)}, \text{ where } (y > 0), \quad f_X(x) = be^{-bx}, \text{ where } (x > 0), \quad a, b > 0.$$

Then, we notice that if α=1, Y would also be Exponential (i.e., the Exponential distribution is a special case of the Gamma with α=1). Now consider Z=X=Y. The PDF is

$$\begin{aligned} f_Z(z) &= \int_{y=0}^{z} f_Y(y) f_X(z-y)\, dy \\ &= \int_{y=0}^{z} \frac{b^{a+1} y^{a-1} e^{-by} e^{-b(z-y)}}{\Gamma(a)}\, dy \\ &= \frac{b^{a+1} e^{-bz}}{\Gamma(a)} \int_{y=0}^{z} y^{a-1}\, dy \\ &= \frac{b^{a+1} e^{-bz}}{\Gamma(a)} \cdot \frac{z^a}{a} = \frac{b^{a+1} z^a e^{-bz}}{\Gamma(a+1)}. \end{aligned}$$

This is a gamma PDF with new shape parameter is $\alpha + 1$ By induction that the sum of n ID exponential variables with common rate parameter λ is gamma with shape parameter α=n rate parameter b=λ.

Therefore by Theorem 4.1 T_n has a Gamma (n,λ) distribution. The Gamma(n,λ) distribution is also more well know as the Erlang distribution, i.e, we can write T_n has an Erlang(n,λ)=Gamma(n,λ), for n = 1,2,3,$\cdots$

The PDF of T_n, for n = 1,2,3,$\cdots$, is given by

$$f_{T_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \quad \text{for } t > 0.$$

Recall that if X has an Exponential(λ), then

$$E[X] = 1/\lambda \quad V[X] = 1/\lambda^2$$

And since T_n= $+X_1$+X_2+$\cdots$+X_n, we concluded that

$$E[T_n] = nE_{X_1} = n/\lambda$$
$$V[T_n] = nV[X_n] = n/\lambda^2$$

Notice that arrival times are not ever going to be independent on each other since that $T_1 < T_2 < T_3 < \cdots < T_n$

Definition 4.2. Inter-Arrival times for Poisson Process
If N(t) has a Poisson Process with a rate λ, then the arrival times T_1, T_2,$\cdots$,T_n have a Gamma (n,λ) distribution. In particular, for n=1, 2, 3, $\cdots$, we have

$$E[Tn] = n/\lambda$$
$$V[Tn] = n/\lambda^2$$

5 Merging and Splitting Processes

Consider N_1(t) and N_2(t) be two independent Poisson processes with rates rates λ_1 and λ_2. Given this information, N(t)= N_1(t)+N_2(t), therefore we can can claim $\lambda = \lambda_1 + \lambda_2$.

Since N1(t) and N2(t) both have independent increments, N(t) also has independent increments. Using the above idea N(t) is a Poisson (($\lambda_1 + \lambda_2$) , t) (the sum of two independent random variables).

Definition 5.1. Merging Independent Poisson Processes
Let N_1(t)+N_2(t),$\cdots$, N_m(t) be m independent Poisson processes with rates λ_1, λ_2,$\cdots$, λ_m. Let also N(t)=N_1(t)+N_2(t),+$\cdots$+N_1(t), for all $t \in [0, \infty)$. Then, N(t) is a Poisson process with rate $\lambda_1 + \lambda_2 + \cdots + \lambda_m$.

Splitting a Poisson process into two independent Poisson processes is very similar to the merging process. Use this example to better understand the topic.

Example 5.1. Suppose that the number of customers visiting a fast food restaurant in a given time interval I is a Poisson distribution with parameter μ denoted as random variable N. Assume that each customer purchases a drink with probability p, independently from other customers, and independently from the value of N. Let X be the number of customers who purchase drinks in that time interval. Also, let Y be the number of customers that do not purchase drinks; so X+Y=N.

1. Find the marginal PMFs of X and Y.

2. Find the joint PMF of X and Y.

3. Are X and Y independent?

Solutions:

1. First, note that the range or X is equivalent to the range of Y (0,1,2,$\cdots$). Also, given N=n, X is a sum of n independent Bernoulli(p) random variables. Thus, given N=n, X has a Binomial distribution with parameters (n,p), so

$$\begin{aligned} X|N=n &\sim Binomial(n,p), \\ Y|N=n &\sim Binomial(n, q=1-p). \end{aligned}$$

$$\begin{aligned} P_X(k) &= \sum_{n=0}^{\infty} P(X=k|N=n)P_N(n) && \text{(law of total probability)} \\ &= \sum_{n=k}^{\infty} \binom{n}{k} p^k q^{n-k} e^{-\mu} \frac{\mu^n}{n!} \\ &= \frac{e^{-\mu}(\mu p)^k}{k!} e^{\mu q} && \text{(Taylor series for } e^x\text{)} \\ &= \frac{e^{-\mu p}(\mu p)^k}{k!}, \qquad \text{for } k=0,1,2,... \end{aligned}$$

   Thus X and Y are both Poisson (μ,q) distributions.

2. To find the joint PMF of X and Y, we use the law of total probability. However, note that P(X=i, Y=j—N=n)=0 if N $\neq$ i+j

$$
\begin{aligned}
P_{XY}(i,j) &= P(X=i, Y=j|N=i+j)P_N(i+j) \\
&= P(X=i|N=i+j)P_N(i+j) \\
&= \binom{i+j}{i} p^i q^j e^{-\mu} \frac{\mu^{i+j}}{(i+j)!} \\
&= \frac{e^{-\mu p}(\mu p)^i}{i!} \cdot \frac{e^{-\mu q}(\mu q)^j}{j!} \\
&= P_X(i)P_Y(j).
\end{aligned}
$$

3. X and Y are independent since, as we saw above,

$$P_{XY}(i,j) = P_X(i)P_Y(j).$$

Definition 5.2. Splitting Independent Poisson Processes
Let N(t) be a Poisson process with rate λ. Here, we divide N(t) to two processes N_1(t) and N_2(t) and for each arrival, there was a half chance that the arrival is sent to the first process (N_1(t)). Otherwise it is sent to the second process. The trials are independent of each other and are independent of N(t). Then,

N_1(t) is a Poisson process with rate λp

N_2(t) is a Poisson process with rate λ (1-p);

N_1(t) and N_2(t) are independent.

6 Practice Exercises and Solutions

6.1

Let $N(t), t \in [0, \infty)$ be a Poisson process with rate λ=0.5.

1. Find the probability of no arrivals in (3,5].
2. Find the probability that there is exactly one arrival in each of the following intervals: (0,1], (1,2], (2,3], and (3,4].

Solutions:

1. If Y is the number arrivals in (3,5], then Y has a Poisson distribution if (μ=1). Therefore,

$$
\begin{aligned}
P(Y=0) &= e^{-1} \\
&= 0.37
\end{aligned}
$$

2. Let Y_1, Y_2, Y_3 and Y_4 be the numbers of independent arrivals in the intervals (0,1], (1,2], (2,3], and (3,4]. Then Y_i has a Poisson(0.5) distribution

$$\begin{aligned} P(Y_1 = 1, Y_2 = 1, Y_3 = 1, Y_4 = 1) &= P(Y_1 = 1) \cdot P(Y_2 = 1) \cdot P(Y_3 = 1) \cdot P(Y_4 = 1) \\ &= \left[0.5e^{-0.5}\right]^4 \\ &\approx 8.5 \times 10^{-3}. \end{aligned}$$

6.2

Let $N(t), t \in [0, \infty)$ be a Poisson process with rate λ, and X_1 be its first arrival time. Show that given N(t)=1, then X_1 is uniformly distributed in (0,t]. That is, show that

$$P(X_1 \leq x | N(t) = 1) = \frac{x}{t}, \quad \text{for } 0 \leq x \leq t.$$

Solution:
For $0 \leq x \leq t$, we can write

$$P(X_1 \leq x | N(t) = 1) = \frac{P(X_1 \leq x, N(t) = 1)}{P\big(N(t) = 1\big)}.$$

$$\begin{aligned} P(X_1 \leq x, N(t) = 1) &= P\Big(\text{one arrival in } (0, x] \quad \text{and} \quad \text{no arrivals in } (x, t]\Big) \\ &= \left[\lambda x e^{-\lambda x}\right] \cdot \left[e^{-\lambda(t-x)}\right] \\ &= \lambda x e^{-\lambda t}. \end{aligned}$$

Therefore, we have

$$P(X_1 \leq x | N(t) = 1) = \frac{x}{t}, \quad \text{for } 0 \leq x \leq t.$$

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